

How does the importance of pre-IPO attributes change across different investment horizons when classifying IPO performance as under- or overperforming the S&P 500

Daniel Garcia, Charles Pearce, Lars Stomps, and Johnny Hu

Erasmus University Rotterdam
Rotterdam, The Netherlands
{637987db, 695309cp, 678137ls, 677926jh}@eur.nl

Abstract. This paper investigates the shift in importance of different pre-IPO attributes when predicting IPO under- or overperformance relative to the S&P 500. We predict short-term (1 day) and long-term (1 Year) horizons using a dataset of 881 IPOs. The variable importances are computed for an Elastic-Net logistic model, a Decision Tree, a Random Forest, and a Gradient Boosting model. Using Feature importance, permutation importance, and drop-one analysis, we see that the importance of different variables shifts significantly between the two horizons. For the 1-day horizon, macroeconomic and IPO characteristic variables are the most important, whereas for the 1-year horizon, firm fundamentals and ownership characteristics are the most important. The level of institutional ownership emerges as a universal predictor of high importance for both horizons and all importance measures.

Keywords: Variable Importance · IPO Prediction · Excess Returns · Elastic-Net Logistic · Decision Tree · Random Forest · Gradient Boosting · 1-day Horizon · 1-year Horizon

1 Introduction

The initial public offering (IPO) is a major milestone for every company. It marks the transition from a private to a public company and allows the firm to raise funds by selling stock. How much the company can raise depends on its valuation, which is estimated using pre-IPO attributes. Many companies try to artificially improve their balance sheet pre IPO to increase their valuation and thus the proceeds collected during their IPO. However, (4) finds that while this leads to higher proceeds in the IPO, it leads to lower subsequent returns, which limits the company's ability to raise capital by issuing equity later on. Another well-known phenomenon in IPO markets is underpricing, which refers to the fact that first-day returns of IPOs are abnormally high. (12) explains that this stems from an underestimation of the correct offer price. If the first day return of

an IPO is high (the IPO is underpriced), the issuing company leaves money on the table, as it could have issued its stock at a higher price and collected more proceeds. (10) shows that on average IPOs leave \$9.1 million on the table, which is a considerable amount for most companies. This can be seen as a cost to the company of going public. Because this *underpricing puzzle* is so expensive, it has drawn a lot of attention.

Many papers investigate the predictability of IPO underpricing. If IPO underpricing can accurately be predicted, the amount of money left on the table can be reduced by adjusting the offer price. In fact, IPO underpricing is fairly predictable with accuracies of 65% to 85% (11), (13). (2) finds that the reputation of the underwriter managing the IPO is an important predictor of IPO underpricing, where more reputable underwriters lead to lower underpricing. (8) investigates IPO underpricing in the Indonesian stock market and finds that the company size and earnings per share are significant predictors of IPO underpricing, where both variables are negatively related to underpricing. (9) shows that pre-IPO analyst coverage and positive market sentiment are valuable predictors of IPO underpricing in the Chinese stock market.

Maximizing IPO proceeds is, however, not the only focus a company should have when going public. (6) find that the median time between the IPO and when a new capital infusion is needed is 12 months. This indicates high importance for the company to retain a high valuation after its IPO so it can raise capital more easily at a later time. Thus, in preparation for its IPO a company must not only focus on increasing its short-term valuation but also on retaining that valuation. However, the research on which variables are important to retain this increased valuation is limited. (1) finds that the 30-day return of an IPO can be predicted with an accuracy of up to 86% using different machine learning algorithms. (5) finds that the industry a company operates in, the age, and the size of the company are significant predictors of the returns 3 years post-IPO, and (3) finds that while the underwriter reputation is an important factor in predicting IPO underpricing (or 1-day returns), its importance in predicting longer-term returns diminishes.

This paper tries to fill this gap in the research by investigating the shift in importance of different variables when predicting first-day IPO returns versus first-year IPO returns in 4 different classification models, where IPOs are classified as either under- or overperforming the S&P 500 index. Knowing how these importances shift can guide the management of a company on where it should focus its attention when preparing for its IPO, such that it can balance high initial proceeds with a retained higher valuation. Additionally, this can be used by investors to make informed decisions when investing in newly listed stocks.

The structure of the remaining part of this paper is as follows. Section 2 provides a detailed description of our data used. Section 3 discusses the implementation of our proposed methods, and the results are evaluated in Section 4. Section 5 concludes this paper and proposes possible additions for future research.

2 Data

Our analysis is based on data combined from three main sources: Bloomberg IPO data, qualitative IPO data from IPOscoop.com (7), and post-IPO performance data scraped from Yahoo Finance. Together, these sources provide a comprehensive view of firm characteristics, market conditions, and both short-term and long-term IPO performance.

The primary dataset was obtained from Bloomberg, containing information on IPOs between 1998 and 2020 with detailed pre-IPO firm characteristics. Specifically, it features financial metrics such as sales growth, profit margin, return on assets, cash flow per share, and offer size; market-related variables including real GDP per capita, CPI growth rate, the OECD Composite Leading Indicator, interest rates, and the seasonally adjusted unemployment rate; as well as firm-level identifiers such as sector, industry group, and industry subgroup. An overview of all features and their definitions is presented in the Appendix in Table 5.

To obtain the first-day returns, we used the dataset provided by IPOscoop.com (7). This dataset facilitated the direct concatenation of the first-day return data with the Bloomberg firm characteristics.

To calculate one-year post-IPO performance, we scraped data from Yahoo Finance using company ticker symbols. Additionally, corresponding S&P 500 returns were collected for each firm at the relevant time horizons. This enabled the computation of both first-day and one-year excess returns relative to the S&P 500 index. The resulting variables, 1st Day Excess Return and 1st Year Excess Return, capture short-term and long-term abnormal IPO performance, respectively.

After integrating the three sources using ticker symbols and IPO dates as matching keys, we obtained a unified dataset of 881 firms with complete coverage across pre-IPO attributes and both short-term and long-term performance outcomes. The combined dataset contains comprehensive financial, structural, and market-related information suitable for analyzing how the importance of pre-IPO characteristics varies across investment horizons. Summary statistics on the excess returns are shown in Table 1.

Table 1: Performance Summary

	n	mean	median	sd	min	max	skew	kurtosis
1 Day Excess Return	881	17.45	8.40	27.84	-100.80	196.27	1.91	5.85
1 Year Excess Return	881	30.82	-10.51	145.51	-117.45	1004.94	3.38	14.83

3 Methodology

This section outlines the classification models, training procedures, and evaluation framework employed to predict binary IPO excess returns. We estimate four machine learning algorithms, selected to capture different complexity levels and provide complementary insights into feature importance. We employ three complementary feature importance analyses (built-in, permutation, and drop-one), alongside a sensitivity and failure analysis. All models are trained using 10-fold Stratified Cross-Validation with class imbalance addressed through SMOTE for 1-day predictions and class weighting for 1-year predictions. Hyperparameter optimization is performed via Grid Search Cross-Validation, and model comparison focuses on out-of-sample accuracy, ROC-AUC, and feature importance rankings across prediction horizons.

3.1 Classification Algorithms

3.1.1 Decision Tree Classifier Decision Trees recursively partition the feature space by selecting at each node the feature j and threshold τ that maximize information gain. For binary classification using Gini impurity:

$$I(t) = 1 - \sum_{k=1}^2 p_k^2(t) \quad (1)$$

where $p_k(t)$ is the proportion of class k at node t . The tree grows until reaching maximum depth or minimum samples per leaf, with predictions determined by majority vote in terminal nodes. We include Decision Trees as an interpretable baseline and to understand which features are used for initial splits.

3.1.2 Random Forest Classifier Random Forest constructs B decision trees, each trained on a bootstrap sample with only $m = \sqrt{p}$ random features considered at each split. The final prediction aggregates individual tree votes:

$$\hat{y} = \text{mode} \{ \hat{y}_b(x) \}_{b=1}^B \quad (2)$$

This ensemble approach reduces overfitting through averaging while maintaining interpretability through feature importance scores aggregated across trees. We use Random Forest to identify robust feature rankings that are stable across different data subsets.

3.1.3 Gradient Boosting Classifier Gradient Boosting sequentially builds trees, with each correcting errors from the previous ensemble. For logistic loss, the algorithm iteratively updates:

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x) \quad (3)$$

where $h_m(x)$ is a tree fit to pseudo-residuals $r_{im} = y_i - p_{m-1}(x_i)$, and $\nu \in (0, 1]$ is the learning rate. Sequential error correction often yields superior accuracy and provides feature importance through cumulative impurity reduction. We include Gradient Boosting to assess whether complex interactions between features improve prediction.

3.1.4 Net-Elastic Logistic Regression For our benchmark model, we will use a logistic regression with an elastic-net regularization. For a binary outcome $y_i \in \{0, 1\}$, where 0 and 1 denote that the IPO is underperforming or overperforming the S&P 500, respectively. Let $i = 1, \dots, n$ be the total number of samples in the dataset and $j = 1, \dots, p$ be the total number of predictors. Then the base model for the logistic regression is the following:

$$p_i = P(y_i = 1 | x_i) = \frac{1}{1 + e^{-(\beta_0 + \sum_{j=1}^p x_{ij} \beta_j)}}$$

where β_0 is the intercept, β_j is the j^{th} predictive variable, and x_{ij} is the value of predictor j for observation i .

We then want to maximize the penalized log-likelihood of this logistic regression.

$$\arg \min_{\beta_0, \beta} \left\{ - \sum_{i=1}^n (y_i \log p_i + (1 - y_i) \log(1 - p_i)) + \lambda \left((1 - \alpha) \frac{1}{2} \sum_{j=1}^p \beta_j^2 + \alpha \sum_{j=1}^p |\beta_j| \right) \right\}$$

where λ is the regularization strength and $\alpha \in [0, 1]$ is the weight of each penalty term. Both the penalty parameter λ and the weight α are computed using 10-fold cross-validation.

3.2 Class Imbalance

The binary classification targets exhibit substantial differences in class distribution. The 1-day excess return target shows severe imbalance with 75.1% outperformers versus 24.9% underperformers (ratio 3.02:1), while the 1-year target is moderately balanced at 43.7% versus 56.3% (ratio 1.29:1). Standard classification algorithms trained on imbalanced data tend to predict the majority class, achieving high accuracy without learning meaningful patterns. We address this through distinct strategies to each horizon.

3.2.1 SMOTE for 1-Day Predictions For the severely imbalanced 1-day target, we employ the Synthetic Minority Over-sampling Technique (SMOTE) (?). SMOTE generates synthetic minority class instances by interpolating between existing observations and their nearest neighbors:

$$x_{new} = x_i + \lambda \cdot (x_{nn} - x_i), \quad \lambda \sim \text{Uniform}(0, 1) \quad (4)$$

where x_i is a minority class instance and x_{nn} is one of its $k = 5$ nearest minority class neighbors. This expands the minority class decision region without duplicating existing observations. Critically, SMOTE is applied only to the training set within each cross-validation fold, never to the test set, preventing data leakage.

3.2.2 Class Weighting for 1-Year Predictions For the 1-year target, we use class weighting rather than oversampling. We employed this to adjust the loss function to penalize minority class misclassifications more heavily:

$$w_0 = \frac{n}{2n_0}, \quad w_1 = \frac{n}{2n_1} \quad (5)$$

where n_0 and n_1 are the number of negative and positive samples. For our 1-year data ($n_0 = 496$, $n_1 = 385$), this yields $w_0 = 0.888$ and $w_1 = 1.144$, meaning underperformer misclassifications receive 28.8% higher penalty. Class weighting preserves the original data distribution while ensuring both classes contribute equally to model optimization.

3.3 Cross-Validation Strategy

To ensure robust generalization and mitigate overfitting, a 10-fold stratified cross-validation procedure was adopted. Stratification preserves the class distribution within each fold, which is crucial for imbalanced datasets such as the 1-day excess return task. For each iteration, nine folds were used for training and one for testing. Data resampling using SMOTE or class weighting was applied exclusively within the training folds to prevent data leakage. The final performance metrics were averaged across all folds to yield stable and unbiased estimates.

3.4 Hyperparameter Optimization

Optimal model configurations were determined using grid search cross-validation, maximizing mean classification accuracy. Each model’s parameter space was designed to balance interpretability and computational efficiency. Decision Trees were optimized over impurity criteria and depth constraints, while Random Forests and Gradient Boosting models were tuned across estimator counts, depth, and learning rate parameters. For the Elastic-Net Logistic Regression, the regularization strength (C) and L1/L2 mixing ratio (α) were jointly optimized. The optimal parameter set θ^* for each model was defined as:

$$\theta^* = \arg \max_{\theta \in \Theta} \frac{1}{K} \sum_{k=1}^K Acc_k(\theta)$$

where $Acc_k(\theta)$ denotes the classification accuracy in fold k , and Θ represents the grid of hyperparameter combinations.

3.5 Feature Importance Analysis

To interpret model behavior and identify key predictors of IPO performance, three complementary approaches were employed.

3.5.1 Built-in Importance. Tree-based models provide intrinsic importance measures based on the average impurity reduction. For feature f_j , its importance is computed as:

$$I_j = \frac{1}{T} \sum_{t=1}^T \sum_{n \in N_{t,j}} p(n) \Delta i(n)$$

where T is the total number of trees, $N_{t,j}$ denotes nodes in tree t that split on feature j , $p(n)$ is the proportion of samples reaching node n , and $\Delta i(n)$ is the impurity decrease from the split.

3.5.2 Permutation Importance. To provide a model-agnostic measure, permutation importance quantifies the change in performance when feature f_j is randomly permuted:

$$I_j^{perm} = A_{baseline} - A_{permuted,j}$$

where $A_{baseline}$ is the model’s original accuracy, and $A_{permuted,j}$ is the accuracy after permuting feature j .

3.5.3 Drop-One Feature Analysis. As a robustness check, each feature was removed individually, and the model retrained to compute the accuracy loss:

$$L_j = A_{all} - A_{-j}$$

where A_{all} is the accuracy with all features and A_{-j} the accuracy without feature j . Large L_j values indicate strong dependency on that feature.

3.6 Sensitivity Analysis

To assess model robustness to perturbations, a sensitivity analysis was conducted by scaling continuous features by small multiplicative factors $p \in \{0.9, 0.95, 1.05, 1.1\}$. For each feature f_j , the mean sensitivity score was computed as:

$$S_j = \frac{1}{|P|} \sum_{p \in P} |A_p - A_{baseline}|$$

where A_p is the model accuracy under perturbation factor p , and $A_{baseline}$ is the unperturbed accuracy. Features exhibiting high S_j values were deemed to contribute to instability, signaling potential overfitting or measurement sensitivity.

3.7 Evaluation Metrics

Given the binary classification framework and the presence of class imbalance, multiple complementary metrics were used to evaluate model performance:

$$\begin{aligned} \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \\ \text{Precision} &= \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \\ F1 &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ \text{ROC-AUC} &= \int_0^1 \text{TPR}(\text{FPR}^{-1}(x)) dx \end{aligned}$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives, respectively. ROC-AUC provides a threshold-independent assessment of model discriminative power, while the F1-score balances precision and recall, especially important for imbalanced data.

3.8 Failure Analysis

To further investigate model limitations, a post-hoc failure analysis was performed. Predictions were categorized as correct, false positives, or false negatives, and associated confidence scores were examined. Comparative feature distributions between correctly and incorrectly classified observations revealed potential biases and data-driven error patterns. This diagnostic analysis helped identify whether model errors arose from intrinsic data noise or systematic misrepresentation.

4 Results

This section presents the empirical results from the classification analysis of IPO excess returns across the two investment horizons. We evaluate the predictive performance of the four machine learning models using 10-fold stratified cross-validation. Performance is reported in terms of accuracy and ROC-AUC, with additional diagnostics for class imbalance, feature importance, and failure patterns.

4.1 Model Performance

Table 2 summarizes the performance of all models across both prediction horizons. Hyperparameter optimization via grid search was performed for all models to maximize accuracy. For 1-day predictions, Gradient Boosting achieved the highest overall accuracy of 70.26% with ROC-AUC of 0.654, outperforming Random Forest (66.86%, AUC 0.659), Logistic Regression (59.14%, AUC 0.604), and Decision Tree (58.80%, AUC 0.590). Although Random Forest achieved a marginally higher

ROC-AUC (0.659), indicating slightly superior class discrimination, Gradient Boosting’s higher accuracy makes it the recommended model for 1-day predictions. For 1-year predictions, Random Forest emerged as the best-performing model with an accuracy of 65.37% and a superior ROC-AUC of 0.691, indicating stronger discrimination between outperforming and underperforming IPOs over longer horizons. Gradient Boosting achieved 63.56% accuracy (AUC 0.674), followed by Elastic-Net Logistic Regression (61.97%, AUC 0.630) and Decision Tree (61.52%, AUC 0.643). The performance gap between models narrows for 1-year predictions compared to 1-day, with all models clustering between 61 and 65 percent accuracy. This compression suggests that long-term outcomes are more difficult to predict consistently, reflecting the increased influence of post-IPO factors not fully captured in pre-IPO attributes.

Table 2: Model Performance Comparison Across Horizons

Model	1-Day		1-Year	
	Accuracy	ROC-AUC	Accuracy	ROC-AUC
Gradient Boosting	70.26%	0.654	63.56%	0.674
Random Forest	66.86%	0.659	65.37%	0.691
Elastic-Net Logistic	59.14%	0.604	61.97%	0.630
Decision Tree	58.80%	0.590	61.52%	0.643

These accuracy levels substantially exceed naive baselines. Random guessing achieves 50 percent accuracy, while always predicting the majority class (outperform) achieves 75.1 percent for 1-day, but learns no discriminative patterns and fails to identify any underperformers. Our models exceed these baselines while maintaining balanced consideration of both classes through SMOTE and class weighting.

The results confirm a strong temporal dimension in predictive accuracy. Short-term outcomes appear more dependent on market timing and sentiment, while long-term results depend more on firm fundamentals. Despite moderate predictive power (65 to 70 percent accuracy), the models capture meaningful structure in IPO performance data, reflecting a level of predictability that exceeds random chance by 15 to 20 percentage points but also highlighting the inherent uncertainty of IPO markets.

4.2 Class Imbalance and Prediction Quality

As discussed in Section 3, the 1-day prediction task was characterized by substantial class imbalance, with 75.1% of IPOs outperforming the market. SMOTE was applied to oversample the minority class. Nevertheless, the Gradient Boosting model displayed a clear imbalance in predictive performance: outperformers were predicted with high accuracy ($F1 = 0.80$) compared to underperformers ($F1 =$

0.38). In contrast, the 1-year model achieved a more balanced performance across both classes ($F1 = 0.68$ for underperformers and 0.57 for outperformers), reflecting a more even class distribution and improved generalization. This asymmetry persists despite SMOTE application, indicating that underperformer patterns in the 1-day data are fundamentally harder to learn, likely because short-term underperformance depends on unpredictable market sentiment and order flow dynamics that leave limited signals in pre-IPO features. A detailed examination of these error patterns is presented in Section 4.5.

4.3 Feature Importance Across Horizons

Feature importance analysis revealed clear differences in the determinants of IPO performance across the two horizons. For 1-day excess returns, the most influential features were largely market and offer related: Offer Price, Institutional Ownership (% Shares Out), Interest Rate, Shares Outstanding, and CPI Growth Rate. These features indicate that short-term outcomes are primarily driven by investor sentiment, macroeconomic context, and IPO pricing mechanics.

In contrast, 1-year excess returns were driven by firm-level fundamentals and institutional involvement. The top features included Institutional Ownership (% Shares Out), Profit Margin, Institutional Owner Shares Held, Unemployment Rate, and Offer Size. Profitability and ownership structure emerge as stronger predictors of sustained performance. These shifts highlight that while early returns are shaped by sentiment and offer dynamics, long-term performance aligns more closely with intrinsic firm quality and investor base composition.

The magnitude of this shift is substantial. Offer Price importance drops from 0.125 (rank 1) for 1-day to 0.033 (rank 8) for 1-year predictions, while Profit Margin rises from 0.061 (rank 6) to 0.119 (rank 2). This quantitative evidence confirms that prediction mechanisms fundamentally differ across horizons rather than simply reweighting the same set of factors.

4.4 Permutation and Drop-One Analyses

To validate the robustness of feature importance rankings, we employed two complementary diagnostic approaches: permutation importance and drop-one analysis. Tables 3 and 4 present the most important variables identified through both methods for 1-day and 1-year returns, respectively.

Permutation importance reinforced the above findings. Interest rates and unemployment dominated short-term importance, while institutional ownership and profitability dominated long-term predictions. Drop-one feature tests provided additional insight, showing that removing Sales Growth (1-Year) led to the largest accuracy decline (4.11 percent) in the 1-year model, underscoring its latent influence. Similarly, excluding Offer Price reduced the 1-day model accuracy by 1.93 percent, confirming its central role in early performance prediction.

Drop-one analysis was performed exclusively on Gradient Boosting models for three strategic reasons. First, computational cost: this method requires retraining 18 times (once per feature) with 10-fold cross-validation, totaling 180 model fits

per algorithm. For all four models, this would require 720 fits. Second, Gradient Boosting achieved the highest 1-day accuracy (70.26 percent) and the second-highest 1-year accuracy (63.56 percent), making it the most relevant model for deployment. Third, we already conduct permutation importance across all models, providing comprehensive feature ranking validation. Drop-one serves as an additional robustness check for our best-performing model rather than a replacement for permutation analysis. The consistency between drop-one (GB only), permutation (all models), and built-in importance (tree models) confirms our feature rankings are not method-dependent.

Table 3: 1-Day Gradient Boosting Variable Importance

Permutation Importance			Drop-One Accuracy Loss		
Feature	Mean	SD	Feature	Loss	% Loss
Interest Rate	0.076	0.005	Offer Price	0.014	1.930
Unemployment Rate	0.066	0.005	Institutional Owner	0.007	0.960
			Shares Held		
Offer Price	0.062	0.006	Industry Group	0.003	0.490
Institutional Owner	0.050	0.004			
% Shares Out					
CPI Growth Rate	0.032	0.004			

Table 4: 1-Year Random Forest Variable Importance

Permutation Importance			Drop-One Accuracy Loss		
Feature	Mean	SD	Feature	Loss	% Loss
Institutional Owner	0.063	0.009	Sales 1-Yr Growth	0.026	4.110
% Shares Out					
Institutional Owner	0.049	0.009	Industry Sector	0.019	3.040
Shares Held					
Return on Assets	0.032	0.005	Real GDP Per	0.014	2.140
			Capita		
Profit Margin	0.031	0.007	Institutional Owner	0.010	1.610
			Shares Held		
Industry Sector	0.031	0.008			

4.5 Failure Analysis

To understand model limitations beyond aggregate performance metrics, we examined prediction errors by classifying outputs as correct predictions, false positives (predicted outperform, actually underperformed), or false negatives

(predicted underperform, actually outperformed). This analysis reveals systematic patterns in where models fail.

For the 1-day Gradient Boosting model, false positives occurred at significantly higher rates than false negatives. Of the 219 IPOs that underperformed the market on day one, 140 were misclassified as outperformers, representing a 64 percent error rate for this class. This bullish bias persists despite SMOTE application and reflects the fundamental challenge of identifying short-term failures in a dataset where 75 percent of IPOs initially outperform. The model’s strong performance on outperformers (recall = 0.82, F1 = 0.80) comes at the cost of poor underperformer detection (recall = 0.36, F1 = 0.38).

The 1-year Gradient Boosting model exhibits a more balanced error distribution, with false positive and false negative rates more evenly distributed across both classes. This reflects the more balanced class distribution (44 percent outperform versus 56 percent underperform) and suggests that longer-term outcomes allow fundamental differences between IPOs to emerge more clearly. The 1-year model achieves F1-scores of 0.68 for underperformers and 0.57 for outperformers, indicating more consistent performance across both prediction targets.

Confidence score analysis shows that prediction probabilities effectively capture uncertainty. Across both models, misclassifications typically occur when predicted probabilities fall in the 0.40 to 0.60 range, creating a gray zone where fundamental and market signals conflict. This uncertain region represents approximately 30 to 35 percent of predictions and indicates cases where model outputs should be interpreted cautiously and combined with qualitative judgment. For practical application, these findings suggest that investors should treat model predictions with predicted probabilities between 0.40 and 0.60 as uncertain, requiring additional due diligence before making investment decisions.

4.6 Interpretation and Discussion

Across all models, institutional ownership consistently ranked among the most informative predictors for both time horizons, suggesting that the participation of institutional investors conveys valuable information about IPO quality and expected performance. This supports the signaling theory in IPO literature, which suggests that informed investor participation reduces information asymmetry.

The temporal shift in feature relevance confirms that different mechanisms drive IPO success across horizons. Short-term outcomes are shaped by market conditions and pricing sentiment, whereas long-term results depend on profitability, sales growth, and capital structure fundamentals.

Overall model performance plateaus around 65 to 70 percent accuracy, indicating a realistic upper bound given the stochastic nature of IPO markets. The ROC AUC scores (0.654 to 0.691) demonstrate that models provide meaningful class discrimination, correctly ranking a random outperformer above a random underperformer 65 to 69 percent of the time. These accuracy levels exceed random guessing by 15 to 20 percentage points, suggesting genuine pattern recognition rather than noise fitting.

However, the performance ceiling also reflects inherent market unpredictability. Post-IPO events such as competitive responses and regulatory changes, management execution quality in implementing business plans, and unforeseen market shocks all shape long-term stock performance but remain unavailable in pre-IPO data. Even with perfect information about pre-IPO attributes, the stochastic nature of business outcomes imposes an upper bound on prediction accuracy. The moderate gap between 1-day (70.26 percent) and 1-year (65.37 percent) accuracy suggests that longer-term outcomes incorporate more post-IPO factors and are therefore inherently more difficult to predict from pre-IPO attributes alone. The larger gap between training and test performance for the 1-day model with SMOTE compared to the 1-year model suggests that synthetic oversampling may introduce slight optimism, though cross-validation mitigates this risk.

Therefore, these models should be interpreted as probabilistic screening tools that reduce but do not eliminate investment uncertainty, serving as aids that complement rather than replace fundamental analysis and human judgment.

5 Conclusion

This paper demonstrates that the predictive importance of pre-IPO attributes fundamentally changes across investment horizons when classifying IPO performance relative to the S&P 500. Using four machine learning classifiers on 881 IPOs, we find that Gradient Boosting achieves the highest 1-day accuracy (70.26%), while Random Forest delivers superior 1-year performance (65.37% accuracy, 0.691 ROC AUC). Feature importance analysis reveals that short-term outcomes are driven primarily by offer mechanics, macroeconomic conditions, and market sentiment with *Offer Price*, *Interest Rate*, and *Unemployment Rate*. In contrast, long-term performance depends more heavily on firm fundamentals and institutional validation, with *Profit Margin*, *Sales Growth*, and *Institutional Ownership* emerging as key predictors. Institutional ownership is the only universal signal, ranking among the top features across both horizons in all three importance analyses (built-in, permutation, and drop-one). Our findings provide meaningful insight for both investors and companies involved in IPOs.

Future research should include more horizons (1 month, 6 months) to test whether this shift happens slowly or suddenly. Additionally, longer horizons could be used to investigate whether this change in variable importance persists and becomes more extreme or if new variables become more important. To further confirm our findings, new models could be trained with only the features found to be important here and test if this significantly improves predictive accuracy.

Data Availability

The dataset and all code used in this study are publicly available at:

<https://github.com/d-bruna/ipo-performance-prediction>

Bibliography

- [1] Alahmadi, M.: A risk-optimized framework for data-driven ipo underperformance prediction in complex financial systems. *Systems* **13**(3), 179 (2025)
- [2] Carter, R., Manaster, S.: Initial public offerings and underwriter reputation. *the Journal of Finance* **45**(4), 1045–1067 (1990)
- [3] Carter, R.B., Dark, F.H., Singh, A.K.: Underwriter reputation, initial returns, and the long-run performance of ipo stocks. *The journal of finance* **53**(1), 285–311 (1998)
- [4] DuCharme, L.L., Malatesta, P.H., Sefcik, S.E.: Earnings management: Ipo valuation and subsequent performance. *Journal of Accounting, Auditing & Finance* **16**(4), 369–396 (2001)
- [5] From, A., Grønkjær, M.B.: Evaluating long-term performance on underpriced ipos (2024)
- [6] Hertzfel, M.G., Huson, M.R., Parrino, R.: Public market staging: The timing of capital infusions in newly public firms. *Journal of Financial Economics* **106**(1), 72–90 (2012)
- [7] IPOscoop: Iposcoop: Ipo data & ratings. <https://www.iposcoop.com/> (nd), accessed October 8, 2025
- [8] Isynewardhana, D., Febryan, F.V.: Factors affecting underpricing level during ipo in indonesia stock exchange 2018-2019. *The Indonesian Accounting Review* **12**(1), 87–98 (2022)
- [9] Jia, C., Ritter, J.R., Xie, Z., Zhang, D.: Pre-ipo analyst coverage: hype or information production? Available at SSRN 3086114 (2018)
- [10] Loughran, T., Ritter, J.R.: Why don't issuers get upset about leaving money on the table in ipos? *The Review of financial studies* **15**(2), 413–444 (2002)
- [11] Quintana, D., Sáez, Y., Isasi, P.: Random forest prediction of ipo underpricing. *Applied Sciences* **7**(6), 636 (2017)
- [12] Ritter, J.R., Welch, I.: A review of ipo activity, pricing, and allocations. *The journal of Finance* **57**(4), 1795–1828 (2002)
- [13] Sonsare, P., Pande, A., Kurve, A., Kumar, S., Shanbhag, C.: A comparative analysis of machine learning algorithms for ipo underperformance prediction. *Journal of Advanced Applied Scientific Research* **5**(6), 1–12 (2023)

A

Table 5: Overview of Features and Definitions

Feature	Definition
Sales Growth (1-Year)	The percent growth of the company's sales in the year before its initial public offering.
Profit Margin	The company's percentage of dollar sales that is profit.
Return on Assets	The company's net income divided by total assets.
Offer Size	The offer price multiplied by the number of common shares sold in the share offering.
Shares Outstanding	The number of shares released to the public.
Offer Price	The price of a single share at the start of the first day it is publicly traded.
Market Cap at Offer	Shares outstanding multiplied by the offer price.
Cash Flow per Share	The company's after-tax earnings plus depreciation divided by the number of shares.
Institutional Ownership (% Shares Out)	The percentage of total shares that are not owned by the company.
Institutional Ownership Shares Held	The number of shares held by institutional investors.
Real GDP per Capita	The value of the final goods and services produced in the United States.
OECD Composite Leading Indicator	The OECD Composite Leading Indicator (CLI) is an index that provides early signals of turning points in economic activity by summarizing a range of forward-looking economic variables.
Interest Rate	The interest rate at which the central bank lends to commercial banks to meet liquidity needs.
Seasonally Adjusted Unem- ployment Rate	The number of unemployed as a percentage of the labor force, adjusted for normal seasonal changes.
CPI Growth Rate	Growth rate from the previous period for the Consumer Price Index.
Industry Sector	The broad industry sector the IPO falls into, represented as a number from 0 to 9.
Industry Group	The group within its sector that the IPO falls into, represented as a number from 0 to 63.
Industry Subgroup	The subgroup within its group that the IPO falls into, represented as a number from 0 to 264.
1st Day Excess Return	The IPO's first-day return in excess of the S&P 500 return over the same period. Captures short-term abnormal performance relative to the market.
1st Year Excess Return	The IPO's one-year return in excess of the S&P 500 return over the same period. Captures long-term abnormal performance relative to the market.